

UTTA-Med

Uncertainty-Aware Test-Time Adaptive Deep Learning
for Reliable Medical Image Classification Under Distribution Shift

Complete Project Explanation
+ 4-Week Team Action Plan

Master Plan v4 — Frozen for Implementation

Primary Dataset	Camelyon17-WILDS
Primary Model	ResNet-18
Core Method	MC-Dropout Uncertainty-Gated TTA
Mandatory Baselines	Source-only + Tent + EATA
Gate Controls	Random / Confidence / Entropy
Statistics	≥5 seeds + WSI-level bootstrap
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This document consolidates the full conceptual foundation and a realistic 4-week execution plan for a research team. All experimental rules from Master Plan v4 are binding.

Part 1 — Complete Project Explanation

1.1 One-Sentence Goal

We want to answer: **Can we make test-time adaptation safer and more reliable for medical image classification under real hospital distribution shift by only adapting on low-uncertainty samples?**

1.2 The Real-World Problem

A deep learning model trained on histopathology patches from Hospitals A, B, C usually performs worse when deployed on Hospital D (different scanner, staining protocol, laboratory, patient population). This is called **distribution shift** (or domain shift).

Formally: $P_{\text{source}}(X, Y) \neq P_{\text{target}}(X, Y)$. Accuracy, AUROC, and especially **confidence calibration** degrade. The model becomes overconfident on wrong predictions.

1.3 Test-Time Adaptation (TTA)

TTA tries to fix this **at deployment time** using only unlabeled target images (no labels from the new hospital):

```
Source hospitals (labeled) → Train model → Freeze checkpoint
↓
Unlabeled target hospital images → Adapt model (usually by minimizing entropy)
↓
Make predictions on target
```

Classic method (Tent): Minimize the entropy of the model's predictions while updating only BatchNorm affine parameters (γ, β). This is simple and strong, but it has a known failure mode: the model can confidently make a **wrong** prediction, then adapt using that wrong signal, and become even worse (error amplification / harmful adaptation).

1.4 Our Core Idea (UTTA-Med)

We estimate how uncertain the model is about each target sample using **Monte Carlo Dropout** (keep dropout active at inference and run multiple forward passes).

If uncertainty $U(x)$ is high → skip the sample (do not adapt on it).

If uncertainty is low → allow the sample to contribute to the entropy-minimization update.

Gate: $w(x) = 1$ if $U(x) < \tau$, else 0

Adaptation loss (only on accepted samples):

$L = \sum(w \cdot H(p)) / \sum w$ (if $\sum w = 0$, skip the batch entirely – no epsilon smoothing)

1.5 Why This Design Is Strong for a Best-Paper Attempt

We do **not** claim “first uncertainty + TTA for medical images”. The literature is too close (EATA, EATA-C, SAR, COME, CertainTTA, etc.). Our defensible contributions are:

- **Reproducible leakage-free Camelyon17-WILDS protocol** for TTA.
- **Hard MC-Dropout predictive-variance gate** (Bayesian-style epistemic uncertainty) instead of point-estimate entropy or confidence.
- **Gate-control ablations** (Random / Confidence / Entropy) + **matched-coverage** protocol → isolates whether the uncertainty signal itself matters.
- **Joint reliability evaluation**: accuracy + calibration (ECE, Brier) + coverage + harmful-adaptation rate + Grad-CAM.
- **WSI-level bootstrap statistics** (patches from the same slide are correlated — patch-level stats are invalid).

1.6 Primary Dataset — Camelyon17-WILDS

- 96×96 RGB patches from lymph-node whole-slide images.
- Binary label: tumor vs non-tumor (based on the central 32×32 region).
- 5 hospitals → natural multi-center shift.

- Official WILDS splits must be used (never re-split yourself).
- **Critical:** Hospital numbering differs across documents → we load the official wilds package, inspect the real domain IDs, and freeze the mapping in `configs/camelion17.yaml`. That file becomes the single source of truth.

1.7 Key Technical Concepts the Whole Team Must Master

Concept	What it is	Why it matters here
Domain / Distribution Shift	Source and target data come from different distributions	The entire problem
Test-Time Adaptation (TTA)	Adapt a frozen model using unlabeled target data at inference	Core technique family
Entropy Minimization (Tent)	Force the model to make confident predictions	Standard TTA baseline
Monte Carlo Dropout	Approximate Bayesian uncertainty by multiple stochastic forward passes	Our uncertainty estimator
Epistemic vs Aleatoric Uncertainty	Model uncertainty vs data noise	We target epistemic uncertainty
Calibration (ECE, Brier)	Does 80% confidence really mean 80% accuracy?	Reliability is as important as accuracy
Harmful Adaptation	Correct prediction becomes wrong after TTA	Central reliability metric
Coverage	Fraction of target samples that pass the gate	Prevents “we just adapted on more data” confound
Matched Coverage	Force competing gates to accept ~same % of samples	Fair comparison
WSI-level Bootstrap	Resample whole slides, not individual patches	Correct statistical inference for histopathology
Grad-CAM	Visualize which regions the model looks at	Explainability audit (not clinical validation)
BatchNorm Affine Only	Update only γ and β of BatchNorm layers	Fair update-scope parity across all methods

1.8 Mandatory Baselines & Controls

ID	Method	Gate / Notes
B1	Source-only (ResNet-18)	None — lower bound
B4	Tent	None — standard TTA
B5	EATA	Method-specific (frozen second baseline)
B6	Random-gated TTA	Random
B7	Confidence-gated TTA	Softmax confidence
B10	Entropy-gated TTA	Predictive entropy
B8	UTTA-Med	MC-Dropout predictive variance (proposed)

All TTA methods update **only BatchNorm affine parameters** (update-scope parity).

1.9 Absolute Rules (Never Break)

1. Target labels are used **only** for final evaluation.
2. τ and all hyperparameters are selected on the OOD validation hospital only.
3. Official WILDS splits only — never re-split.
4. No epsilon smoothing for empty gates — explicitly skip the batch.
5. Report coverage for every gated method.
6. Primary statistics = WSI-level bootstrap.
7. Minimum 5 seeds (fixed list: 42, 123, 2024, 7, 99).

Violation of any of the above invalidates the scientific claims of the paper.

1.10 What Counts as Success

We do **not** require UTTA-Med to win every metric. Positive, mixed, or negative results are all scientifically valid if the experiments are rigorous and honestly reported. The paper wins by clarity of analysis, not by forced superiority.

Part 2 — Complete 4-Week Team Action Plan

Assumptions: Team of 3–5 people; one strong PyTorch lead; one data/statistics lead; one evaluation/results lead; shared Git repo (already initialized); at least one GPU (8–16 GB VRAM recommended; can start on CPU/subset).

Week 1 — Foundation, Data, Source Model, Shift Demonstration

Goal: Everything needed to prove the problem exists and the pipeline is leakage-free.

Day	Focus	Deliverables	Owner
1–2	Infrastructure	Python 3.10+ env, pip install -r requirements.txt, Git workflow, basic tracking helper, dummy run logged	Infra lead
2–3	Data Layer	PathMNIST + PCam loaders; Camelyon17 official wilds loader; inspect real domain IDs and freeze configs/camelyon17.yaml	Data lead
3	Leakage Tests	tests/test_no_leakage.py passes (WSI intersection = empty)	Data lead
3–4	Dataset Exploration	Class balance, hospital/WSI distributions, sample images → results/data/	Data lead
4–5	Source Model	ResNet-18 + classification head, source-only training, source metrics	Model lead
5–6	Distribution Shift	Target evaluation of frozen source model; Drop metrics; optional UMAP/t-SNE	All
6–7	Buffer + Review	Fix any leakage or split issues; Stage 0–5 acceptance gates passed	Team review

Week 1 Exit Gate: Source model trains, official splits verified, no leakage, clear performance drop on target hospital documented.

Week 2 — Standard TTA + Uncertainty Estimation

Goal: Working Tent + EATA + reliable MC-Dropout uncertainty.

Day	Focus	Deliverables
1–2	Tent	Correct BN-affine-only Tent implementation; same update scope as all future methods
2–3	EATA	EATA baseline (restrict to BN-affine if needed); document any deviation
3–4	MC-Dropout	Uncertainty head / dropout kept active; N-pass predictive variance; sensitivity for $N = 5, 10, 20, 30, 50$
4–5	Uncertainty Validation	Bin uncertainty → error rate plot (must show $U \uparrow \Rightarrow \text{Error} \uparrow$). Do not proceed to gating if signal is weak
5–6	Basic Adaptation Loop	Shared adaptation engine that all gated methods will reuse
6–7	Buffer + Code Review	Clean, tested Tent / EATA / MC-Dropout; uncertainty–error relationship confirmed

Week 2 Exit Gate: Tent and EATA improve over source-only (or the failure is understood); MC-Dropout uncertainty correlates with error.

Week 3 — UTTA-Med Core + Gate Controls + Reliability

Goal: Full proposed method + all controls + calibration + harmful-adaptation analysis.

Day	Focus	Deliverables
1	UTTA-Med Gate	Hard gate + zero-accepted-batch skip; τ sweep on validation only
1–2	Gate Controls	Random, Confidence, Entropy gates with matched-coverage protocol
2–3	Harmful Adaptation	Before/After vs Ground-Truth analysis; Correction Rate + Harm Rate
3–4	Calibration	ECE, Brier, reliability diagrams for every method
4–5	τ Sensitivity + Stability	Full τ curves; adaptation-step sweep (1/2/5/10)
5–6	Ablation Matrix	Complete A–F table (including B10 Entropy)

Day	Focus	Deliverables
6–7	Multi-seed Dry Run	At least 2–3 seeds on the main matrix to catch bugs

Week 3 Exit Gate: UTTA-Med end-to-end works; all gate controls run; coverage, harm rate, ECE reported; τ frozen from validation.

Week 4 — Full Experiments, Statistics, Figures, Paper Skeleton

Goal: Publication-ready results + first complete paper draft.

Day	Focus	Deliverables
1–3	Full Multi-seed Matrix	≥ 5 seeds on all mandatory methods (B1, B4, B5, B6, B7, B10, B8)
2–3	WSI-level Statistics	<code>bootstrap_ci_by_slide()</code> implemented and used for all headline CIs
3–4	Grad-CAM	Correct / Incorrect / High-U / Wrong→Correct / Correct→Wrong examples
4	Figures & Tables	All required figures + main results table populated with real numbers
4–5	Reproducibility Audit	Every number in tables traceable to experiment ID + Git commit + config
5–6	Paper Skeleton	Introduction, Related Work (explicit EATA/SAR differentiation), Method, Experiments, Ablation, Discussion, Limitations
6–7	Final Review + Buffer	Team walkthrough of results honesty; optional MedMNIST-C if time remains

Week 4 Exit Gate: All mandatory experiments finished, statistics correct, figures ready, first full paper draft written, repository publicly releasable.

Team Role Suggestions (Flexible)

- **Data & Protocol Lead:** Camelyon17 mapping, leakage tests, WSI bootstrap, statistical analysis.
- **Core Method Lead:** ResNet, Tent, EATA, MC-Dropout, UTTA-Med gate, adaptation engine.
- **Evaluation & Results Lead:** Metrics, calibration, harmful adaptation, Grad-CAM, tables, figures.
- **Paper Lead:** Related Work positioning, writing, reproducibility checklist.
- Everyone shares experiment running and debugging.

Risk Buffer Rules

If something slips, **never cut:** leakage control, matched coverage, gate controls, WSI-level stats, 5 seeds, target-label isolation.

Cut first: ResNet-50, SAR, DLTTA, EATA-C, MedMNIST-C, temperature scaling, 10-seed extension.

Daily Stand-up Suggestion (15 min)

1. What gate did we pass yesterday?
2. What is blocked today?
3. Any risk of target-label leakage or statistical incorrectness?

Final Status

Master Plan v4 is frozen. Repository Stage 0 is complete (docs, configs, Git). The team can begin Week 1 Day 1 immediately: environment setup → official WILDS loader inspection → freeze configs/camelyon17.yaml → leakage tests → source model.

Negative or mixed experimental results remain scientifically valid and must be reported honestly. The scientific objective is to establish whether MC-Dropout predictive uncertainty provides useful information for safer and more reliable TTA — not to force UTTA-Med to win every metric.

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